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Problem Solving Protocol

Bridging-BPs: a novel approach to predict potential drug–target interactions based on a bridging heterogeneous graph and BPs2vec

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Abstract

Predicting drug–target interactions (DTIs) is a convenient strategy for drug discovery. Although various computational methods have been put forward in recent years, DTIs prediction is still a challenging task. In this paper, based on indirect prior information (we term them as mediators), we proposed a new model, called Bridging-BPs (bridging paths), for DTIs prediction. Specifically, we regarded linkage process between mediators and DTs (drugs and proteins) as 'bridging' and source (drug)-mediators-destination (protein) as bridging paths. By integrating various bridging paths, we constructed a bridging heterogeneous graph for DTIs. After that, an improved graph-embedding algorithm—BPs2vec—was designed to capture deep topological features underlying the bridging graph, thereby obtaining the low-dimensional node vector representations. Then, the vector representations were fed into a Random Forest classifier to train and score the probability, outputting the final classification results for potential DTIs. Under 5-fold cross validation, our method obtained AUPR of 88.97% and AUC of 88.63%, suggesting that Bridging-BPs could effectively mine the link relationships hidden in indirect prior information and it significantly improved the accuracy and robustness of DTIs prediction without direct prior information. Finally, we confirmed the practical prediction ability of Bridging-BPs by case studies.

Keywords: drug–target interaction, mediators, bridging heterogeneous graph, BPs2vec

Introduction

Drugs can interact with multiple targets, which is the reason why drugs exhibit multiple pharmacological properties and drug side effects [1–3]. Instructively, identifying novel efficacy for original drugs by exploring drug–target interactions (DTIs) is a key strategy for repurposing known drugs [4]. However, although many high-throughput screening and bioassays are emerging, experimental methods for DTIs prediction are usually laborious and costly [5, 6]. It is of great significance to propose effective computational methods to predict potential DTIs.

Docking simulation methods [7] and ligand-based methods [8] are early attempts at computational DTIs prediction. For the docking-based approaches, the three-dimensional structures of proteins and targets are considered to identify potential binding sites, which results in a time-consuming prediction process with failure risk to identify DTIs due to lack of three-dimensional structure information. Ligand-based approaches infer potential ligands by analyzing known ligands and target proteins and may perform poorly in those scenarios with fewer known ligands.

Recently, public databases involved in DTIs have been released. Various graph/network-based DTIs prediction methods have been proposed [9, 10]. For instance, MTOI [11] inferred potential drug targets by systematically analyzing the transformation between the disease state and the desired state in a disease network. Besides, similarity networks, such as drug side-effect similarity [12] and multiple similarity subgraph fusion [13] have been introduced in DTIs prediction. Despite their successes, above methods rely on direct prior information (i.e. known DTIs) to some extent. Once removing most of known DTIs, their model might not work well. Machine learning-based methods are more widely applied in DTIs prediction. These methods try to integrate more different types of biological data (e.g. association information, similar-

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ity information and sequence structure information of biomedical entities) to improve performance [14–17].

In 2019, Molecular Associations Network (MAN) [18] have been proposed. It provides a global idea for detecting DTIs. In MAN, various molecules and corresponding associations are regarded as nodes and edges in a heterogeneous network, which indicates that MAN is a complex network instead of simple bipartite network. To identify potential associations for pair-wise molecule in MAN, network representation learning (NRL) methods, such as node2vec, GraRep, Large-scale Information Network Embedding (LINE) are used to obtain low-dimensional node embeddings [19–21]. MAN-based prediction methods have promising performance for miRNA-disease and lncRNA-disease association prediction [22–24]. However, these methods still existed some deficiencies: First, they rely much on direct prior information. In general, less reliable direct prior information verified by experiments could be obtained. Once we use the sparse direct prior information to train prediction model, the model performance is unreliable. Second, they fail to take into account those typical subgraph structures benefited for uncovering DTIs in MAN.

Inspired by MAN, we argue that mediators (i.e. indirect prior information) are conducive to DTIs prediction. It is expected to link mediators to isolated drug and target nodes for DTIs prediction with less direct prior information. Here, we specifically regard the linkage process between mediators and DTs (proteins and diseases) as 'bridging', and source (drug)-mediators-destination (protein) as bridging paths (BPs). We mapped the potential multi-hop links between DTs into BPs. To obtain more bridging information, it is a feasible strategy to synthesize various bridging paths to build a bridging heterogeneous graph for DTIs. Essentially, a bridging heterogeneous graph is consisted of various types of BPs. Node feature representations can be captured more precisely by calculating contribution value of each subgraph derived from BPs. Therefore, we proposed a BPs-biased walking algorithm, BPs2vec, to carry out biased walks over bridging heterogeneous graph. Its core idea are as follows: (i) taking BPs to divide the bridging heterogeneous graph into multiple subgraphs; (ii) calculating the Path Contribution Factor (PCF, its definition can be found in method section) of each subgraph; (iii) performing BPs-biased walk to obtain walking sequences of each subgraph and (iv) incorporating these walking sequences with different subgraph structures to obtain final sequence sets according to corresponding PCF.

In this paper, we proposed a new DTIs prediction model, Bridging-BPs, based on indirect prior information. Specifically, we first screened three types of mediators including lncRNA, miRNA and disease to form various types of BPs, where multiple types of mediators can make various reasonable bridging paths. Second, by virtue of these BPs, we established a bridging heterogeneous graph. Then, we employed BPs2vec to capture deep topological features among nodes in the bridging heterogeneous graph, thus to obtain the low-dimensional node vector representations. Finally, the vector representations were fed into a Random Forest (RF) classifier for non-end-to-end training, outputting final DTIs prediction results. Under 5-fold cross validation, Bridging-BPs obtained AUPR of 88.97% and AUC of 88.63%, indicating Bridging-BPs could effectively mine the link relationships hidden in the indirect prior information. Comprehensive experiment results demonstrated the accuracy and robustness of Bridging-BPs without direct prior information. Case studies also confirmed its practical prediction ability for new DTIs.

Materials and methods

Problem and model description

In this work, the set of DTIs is represented as a tuple (DTs, Ac). DTs is represented a set consisted of drugs D and target proteins T, where $D=\{d_1, d_2, \ldots, d_n\}$ and $T=\{t_1, t_2, \ldots, t_k\}$. Ac denotes associations among DTs, which is defined as: $Ac=\{ac(d_1, t_1), ac(d_2, t_2), \ldots, ac(d_n, t_k)\}$. If d_n interacts with t_k, then $ac(d_n, t_k)$=1, if not then $ac(d_n, t_k)$=0. We represent the vector feature $f_{(d_n,t_k)}$ of (d_n, t_k) as the concatenating of structural feature f_{d_n} of d_n and structural feature f_{t_k} of t_k. In downstream classification task, a RF classifier can obtain final prediction result by scoring $f_{(d_n,t_k)}$, which can be represented as $ac(d_n, t_k)=\mathrm{RF}(f_{(d_n,t_k)})$.

It is worth emphasizing that we aim to identify potential DTIs using indirect prior information based on the bridging heterogeneous graph and BPs2vec. So that, we constructed a bridging heterogeneous graph to integrate different bridging paths. Moreover, to make full use of topology information of the graph, we proposed an improved NRL method, BPs2vec and trained DTs vector representation obtained by BPs2vec using a RF classifier to get the final classification result. The workflow of Bridging-BPs is shown in Figure 1.

The bridging heterogeneous graph construction

To obtain more comprehensive bridging information, we selected three types of mediators (miRNA, lncRNA and disease) to construct BPs. Even so, the multiplicity of mediator-related associations greatly increases the diversity of BPs, suggesting the necessity of constructing a high-quality bridging heterogeneous graph $G(V, E)$, where V denotes the set of nodes, including source nodes, mediators and destination nodes; E denotes the set of edges (i.e. mediator-related associations).

Specifically, we downloaded eight types of indirect prior information, including miRNA-lncRNA associations, miRNA-protein associations, miRNA-disease associations, lncRNA-disease associations, lncRNA-protein associations, drug–disease associations, disease-protein associations and protein–protein interactions from eight databases [25–32]. Meanwhile, to verify the effectiveness of our model, we download 5464 pairs of known DTIs from DrugBank [33] as test dataset. The data are shown in Tables 1 and 2.

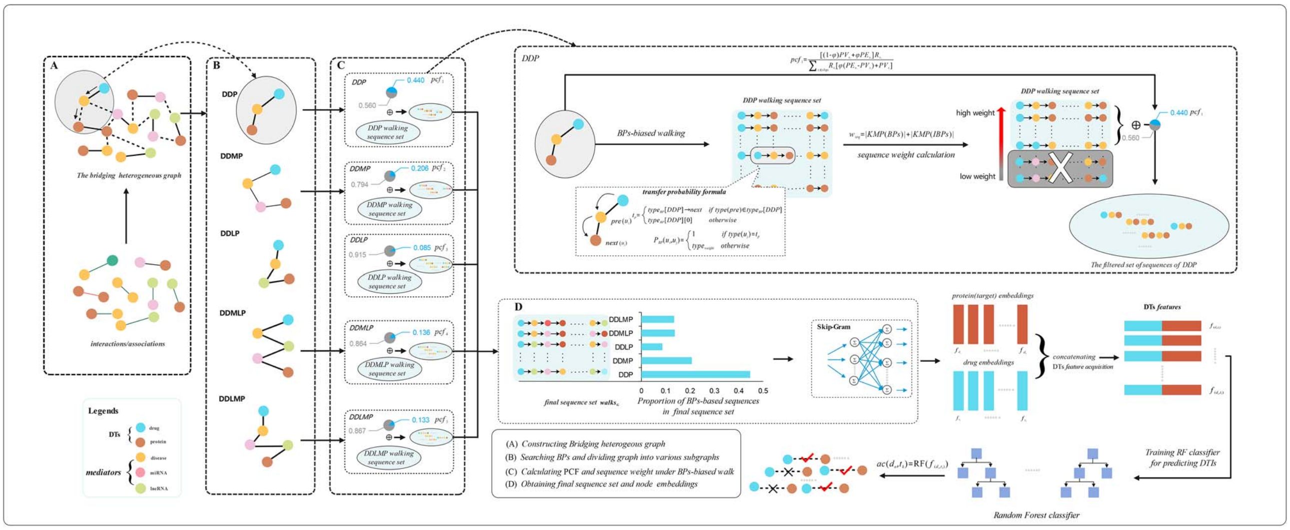

Figure 1. The workflow of Bridging-BPs. Bridging-BPs consists of five steps including: (A) By integrating various BPs, we constructed a bridging heterogeneous graph for DTIs. (B) We performed BPs Searching in the bridging heterogeneous graph to select BPs and performed graph division based on BPs. (C) We calculated PCF of each subgraph. After that, we performed BPs-biased walk on different subgraphs to obtain various walking sequences. Meanwhile, weight operations were performed on sequence sets. (D) Based on PCF, the weighted walking sequence sets were recombined into a final walking sequence set. Skip-Gram model was applied to get node embeddings. A Random Forest classifier was applied to train node features obtained by BPs2vec and output classification results. Where (B), (C) and (D) belong to the work content of BPs2vec.

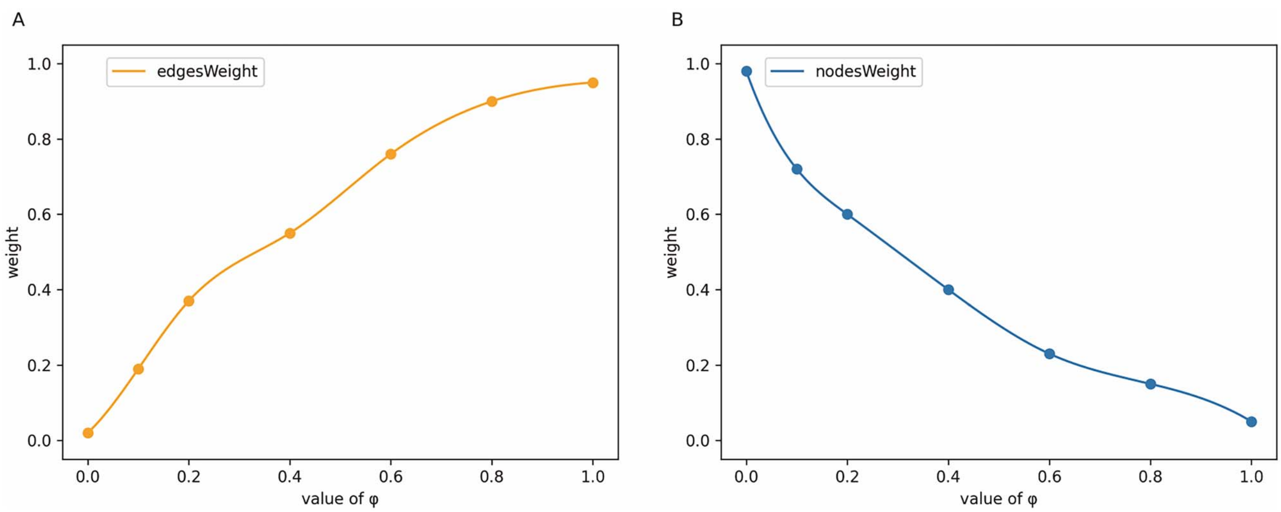

Figure 2. The relationship among φ, the weight of edges (A) and the weight of nodes (B). The sum weight of edges and nodes with same φ is ∼1. When the value of φ is adjusted, the weight of the edge set and the weight of the node set also change in the opposite way.

Table 1. indirect prior information in the heterogeneous graph and the test dataset

Association	Database	Quantity
miRNA–lncRNA	lncRNASNP2	8374
miRNA–protein	miRTarBase	4944
miRNA–disease	HMDD v3.0	16 427
lncRNA–disease	LncRNADisease and lncRNASNP2	1264
lncRNA–protein	LncRNA2Target v2.0	690
Drug–disease	CTD: update 2019	18 416
Disease–protein	DisGeNET	25 087
Protein–protein	STRING: in 2021	19 237
Drug–protein	DrugBank v5.0	5464

BPs2vec

Inspired by node2vec, we proposed a novel-embedding algorithm based on BPs-biased walk to learn the structural information of a bridging heterogeneous graph. The main work of BPs2vec includes: (i) BPs searching and subgraph division; (ii) PCF calculation and

Table 2. Nodes in the bridging heterogeneous graph

Node	Quantity
Disease	2062
Protein	1649
miRNA	1023
Drug	1025
lncRNA	769
Total	6528

Table 3. Subgraphs obtained through graph division

Subgraph	BPs
DDP	Drug–disease–protein
DDMP	Drug–disease–miRNA–protein
DDLP	Drug–disease–lncRNA–protein
DDMLP	Drug–disease–miRNA–lncRNA–protein
DDLMP	Drug–disease–lncRNA–miRNA–protein

(iii) BPs-biased walk and feature acquisition. The step-by-step pseudo-code (Algorithm 1) of BPs2vec is shown as follows:

BPs searching and graph division

To find all BPs in G for BPs2vec, we performed BPs searching over the bridging graph. Specifically, we first traversed and searched all eight types of associations to select fixed mediators including miRNA, lncRNA and disease. Then, we constrained consistency of source and destination nodes in selected paths (i.e. drug for source and protein for destination) thus to obtain the BPs required by BPs2vec. After BPs searching, we found five kinds of BPs. We divided G into five subgraphs derived from above BPs. Subgraphs and corresponding BPs are shown in Table 3.

Path contribution factor calculation

Compared with MAN, we regarded BPs as the smallest functional structure in bridging heterogeneous graphs. Therefore, subgraphs divided according to BPs are functionally specific for DTIs prediction. To quantify functional specificity of subgraphs, we calculated PCF of each subgraph according to its structure. For example, for the subgraph s_1, its PCF pcf_1 is:

$pcf_1 = \frac{[(1-\varphi)PV_{s_1}+\varphi PE_{s_1}]R_{s_1}}{\sum_{s_i \epsilon sbgs} R_{s_i}[\varphi(PE_{s_i}-PV_{s_i})+PV_{s_i}]}$ $\varphi\epsilon[0,1]$ (1).

where PV_{s_1} and PE_{s_1} represent the proportion of nodes and edges of s_1 and G, respectively. PV_{s_1}, PE_{s_1} and R_{s_i} are described as follows:

$$\begin{cases} PV_{s_1} = \frac{|V_{s_1}|}{|V|} rv_{s_1} \; rv_{s_1} = \frac{|NV|}{|NV_{s_1}|} \\ PE_{s_1} = \frac{|E_{s_1}|}{|E|} re_{s_1} \; re_{s_1} = \frac{|NE|}{|NE_{s_1}|} \\ R_{s_i} = \frac{|DT_{s_1}|}{\sum_{s_i \epsilon sbgs} |DT_{s_i}|} \end{cases} \quad (2)$$

where $|\bullet|$ represents a quantity calculation operation. Taking PV_{s_1} as an example, V represents nodes in G and V_{s_1} represents nodes in s_1. The node type ratio rv_{s_1} is applied to balance the weight of nodes types in s_1, where NV represents node types set in G and NV_{s_1} represents node types set in s_1. DT_{s_i} refers to intersections between subgraph s_i and the test dataset at nodes. R_{s_i} is defined to describe the role of DTs intersection in DTIs prediction. φ is a hyper-parameter utilized to regulate PCF for imbalance scenario of edges and nodes. By adjusting φ, the weight of nodes and edges can be restrained. With a complete graph reference standard, we set the value of φ as:

$$\varphi = \begin{cases} range\,[0, 0.5) \;\; if \;\; |NE_{s_1}| \le ne_{avg} \\ range\,[0.5, 1] \;\; otherwise \end{cases} \quad (3)$$

where $ne_{avg}=average(|NV_{s_1}|-1, \frac{|NV_{s_1}|^2-|NV_{s_1}|}{2})$. We obtained PCF distribution at different values of φ, as shown in Figure 2.

BPs-biased walk and feature acquisition

BPs reflect the topological features of nodes in subgraphs. Therefore, we add BPs bias to walk processes performed over each subgraph when sampling, which makes walking sequences contain as many complete BPs as possible. To implement BPs-biased walk, we introduced a Node Type Constant t_p to record the next node type in transfer process. We define t_p as:

$$t_p = \begin{cases} type_{BP}[s_i] \rightarrow next \quad if \; type(pre) \in type_{BP}[s_i] \\ type_{BP}[s_i]\,[0] \quad otherwise \end{cases} \quad (4)$$

where $type(pre)$ denotes the current node type, and $type_{BP}[s_i]$ denotes the set of node types in subgraph $[s_i]$. We obtained a new formula for the transfer probability from node u_i to node u_j:

$$P_{BP}(u_i, u_j) = \begin{cases} 1 \; if \; type(u_j) = t_p \\ type_{weight} \; otherwise \end{cases} \quad (5)$$

where $type_{weight}$ denotes node transfer probability constant, which indicates the transfer probability when the node type does not belong to corresponding BPs.

Besides, we recombine the high-weight sequence subset of each subgraph according to PCF to obtain final sequence set $walks_G$ as:

$$walks_G = cup_{S_i \epsilon sbgs} walks_i \oplus pcf_i \quad (6)$$

where $walks_i$ represents walking sequence set obtained from subgraph s_i, and $\oplus$ denotes the operation of selecting high-weight sequence subsets of $walks_i$ with pcf_i proportion. Specifically, in $\oplus$, the number of subsequences matching BPs is the main characteristic that distinguish the importance of each sampling sequences. In addition,

Algorithm 1 The *BPs2vec* algorithm

LearnFeatures(Graph $G=(V,E,W)$, Dimensions d, Walks per node r, Walk length l, Context size k, TypeWeight tw, TypeSource ts, TypeDestination td, EdgeNodeWeight φ)

π =PreprocessModifiedWeightsNew(G,tw)
$G'=(V,E,\pi)$
$type_{BP}$=BPsSearching (G',ts,td)
$subgs$ = GraphDivision($G',type_{BP}$)
PCF = ContributionCalculation($subgs$, φ)
Initialize *walks* to Empty
for $iter=1$ **to** r **do**
  for $s(V',E') \in subgs$ **do**
    num = $subgs$.index($s(V',E')$)
    for all nodes $u \in V'$ **do**
      $walk$ = BPs2vecWalk($s,u,l,type_{BP}$)
    Append *walk* to $walks[num]$
for $iter$=1 **to** len($subgs$) **do**
  $sbwalks$= shuffle ($walks[iter], PCF[iter]$)
Apped *sbwalks* to *finalWalks*
f = StochasticGradientDescent($k,d,finalwalks$)
return f

BPs2vecWalk(Graph $G'=(V,E,\pi)$, Start node u, Length l, TYPE$_{BP}$ $type_{BP}$)

  Initialize *walk* to $[u]$
  for $walk\_iter=1$ **to** l **do**
    $curr = walk[-1]$
    $type_{next}$ = TypeSelect($type_{BP}, type_{curr}$)
    V_{curr} = GetNeighborsBP(curr, $type_{next}$, G')
    s = AliasSample(V_{curr}, π)
    Append s to *walk*
  return *walk*

.

we argue that an inverse BPs (e.g. the inverse path of drug–disease–protein is protein–disease–drug) is another distinguishing characteristic. we define sequence weight W_{seq} under each subgraph derived from BPs:

$$W_{seq} = |KMP(BPs)| + |KMP(IBPs)| \tag{7}$$

where, *IBPs* denoted inverse BPs. In this formula, we employ Knuth–Morris–Pratt (KMP) Algorithm to calculate BPs-related frequency. Finally, $walks_G$ was fed to Skip–Gram model and BPs2vec eventually got structural features of DTs.

Based on our improved sampling mechanism, we maximize the observation neighborhood of the node u as:

$$\max_w \sum_{u\in V} logPr\left(N_{path}(u)|W_{path}(u)\right) \tag{8}$$

where W: $V_{path} \Rightarrow \mathbb{R}^{64}$ denotes a map function, and W is a matrix of $|V| \times 64$. $N_{path}(u)$ denotes the neighborhood nodes of u obtained by BPs-biased sampling. Similar to node2vec, we adopt conditional independent assumption and characteristic space symmetry assumption, so that we get final target function of BPs2vec:

$$\max_W \sum_{u\in V_{m-path}} \left[-\log Z_u + \sum_{n_i \in N_{path}(u)} W_{path}(n_i) \bullet W_{path}(u) \right] \tag{9}$$

where Z_u is the normalization factor,which was define as:

$$Z_u = \sum_{n_i \in N_{path}(u)} \exp W_{path}(n_i) \bullet W_{path}(u) \tag{10}$$

Training Random Forest for predicting DTIs

After obtaining structural features, we trained a RF classifier for predicting novel DTIs. RF is an integrated learning algorithm that completes prediction by integrating multiple decision trees. Based on bootstrap resampling technique, k samples are randomly selected from the original training sample set N to generate a new training sample set. A RF classifier consists of k classification trees based on bootstrap sample set. And, the number of scores cast by the classification tree determine the classification results of DTIs.

Experimental settings

For a binary classification problem, its prediction states can be divided into the four categories: true positive (TP), false positive (FP), true negative (TN) and false negative (FN). Thus, we select some predictive indicators to evaluate the prediction effect, including precision, recall and specificity values:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{11}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{12}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{13}$$

Based on these evaluation metrics, we construct the receiver-operating characteristic (ROC) curve and precision-recall (PR) curve of prediction results. Meanwhile, we obtain area under the PR curve (AUPR) and area under the ROC curve (AUC). In addition, we select three other evaluation indicators to comprehensively evaluate prediction results, including Accuracy (Acc.), Matthews Correlation Coefficient (MCC) and Sensitivity (Sen.).

Due to the longer the BP, the lower the existing possibility of biological association between source nodes and destination nodes [26], we add up to three types of mediators in BPs. Furthermore, to facilitate comparison of experimental results, we set φ as 0.5 in PCF calculation. Meanwhile, on the setting of the transfer probability, we set the value of the $type_{weight}$ as 0.25.

Moreover, to illustrate the consistency between PCF and the actual contribution of each subgraph in current data set, we convert prediction results of each subgraph s_i into corresponding proportion, and the transformation formula for subgraph s_1 is as follows:$Re\_pt =$

$$\frac{T_{s_1} + \max(T_{subgs})}{\sum_{s_i \in subgs}(T_{s_i} + \max(T_{subgs}))} \tag{14}$$

where T_{s_1} is calculated as:

$$\begin{cases} T_{s_1} = (TP_{s_1} + TN_{s_1}) - \overline{T} \\ \overline{T} = \frac{\sum_i^n (TP_{s_i} + TN_{s_i})}{n} \end{cases} \tag{15}$$

In the formula, TP_{s_1} represents the number of true positive DTIs, and TN_{s_1} represents the number of true negative DTIs. $\overline{T}$ denotes the mean number of successful prediction. Moreover, we introduce a maximum $\max(T_{subgs})$ to avoid negative numbers. As shown in Table 4 and Figure 3, PCF of each subgraph basically matches the actual contribution.

Results

Evaluation of Bridging-BPs

We employed 5-fold cross validation on known DTIs to evaluate the performance of Bridging-BPs. Negative samples were constructed with 5464 known DTIs equal proportions. Training set and test set were taken from the set of positive and negative samples. To reduce the effect of potential sample division bias, we randomly divided positive and negative samples into 5-folds. In each round of Bridging-BPs, 1-fold was processed as test samples and the remaining 4-folds were processed as training samples. The number of validation test samples per round was ∼2186. We recorded evaluation indicators of DTIs prediction. Moreover, to ensure the reliability of evaluation indicators, we calculated mean indicators and their standard deviation. Based on these experimental data, we obtained the PR and ROC curves of our modes. The results are shown in Figure 4 and Table 5.

In fact, the ROC curves show that our model has a reasonable AUC value (88.63%), which demonstrates that a significant number of DTIs in test samples are correctly predicted. In each round of Bridging-BPs, an average of 1772 DTIs is correctly predicted. In addition, it is noteworthy that the number of correct predictions is relatively stable, which makes a small standard deviation for various metrics including AUC and AUPR.

Moreover, PR curves indicate that Bridging-BPs has a high-recall rate. In fact, Bridging-BPs utilized various BPs to capture low-dimensional structural features of nodes, which makes DTs feature representations more comprehensive. Therefore, the predictive result of Bridging-BPs is

Table 4. PCF and *Re_pt* of subgraph s_i

Subgraph	V_{s_i}	E_{s_i}	NV_{s_i}	NE_{s_i}	DT_{s_i}	TN + TP	PCF	*Re_pt*
DDP	2249	43 503	3	2	435	1598	0.440	0.400
DDMP	3117	39 787	4	3	285	1484	0.206	0.165
DDLP	1637	20 370	4	3	227	1472	0.085	0.140
DDMLP	3253	43 907	5	4	227	1468	0.136	0.131
DDLMP	2717	32 998	5	4	285	1483	0.133	0.163

Table 5. The main experiment results under 5-fold cross-validation

Fold	Acc. (%)	Sen. (%)	Spec. (%)	Prec. (%)	MCC (%)	AUC (%)
0	81.84	82.43	81.24	81.46	63.68	88.19
1	82.07	82.89	81.24	81.55	64.14	88.59
2	81.29	83.17	79.41	80.16	62.62	88.62
3	82.39	83.17	81.61	81.89	64.78	88.85
4	82.33	84.16	80.49	81.18	64.70	88.88
Mean	81.98±0.45	83.16±0.63	80.80±0.88	81.25±0.66	63.98±0.88	88.63±0.27

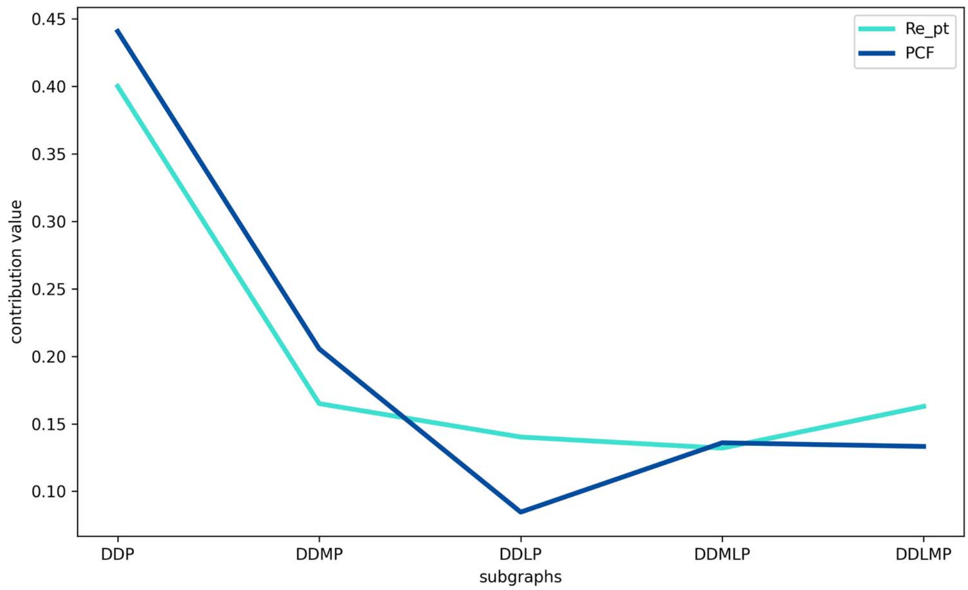

Figure 3. Comparison of PCF and prediction results. PCF and prediction performance of each subgraph have similar trends, which indicates that PCF reflects the expected contribution of subgraphs in DTIs prediction process to some extent.

not susceptible to be affected by the number of positive and negative samples thus to get higher AUPR.

Comparison of different subgraphs

Due to functional specificity of subgraphs derived from BPs, we argued that each subgraph could also be applied to predict potential DTIs to some extent. To illustrate the prediction effect of a single subgraph, we extracted node features in each subgraph by BPs2vec and fed them into a RF classifier. The prediction results are shown in Figure 5 and Table 6. The results show that AUC of each subgraph in DTIs prediction exceed 68%, which indicates that a single subgraph can reflect structural features of nodes to some extent. Table 6 shows that recombining walking sequences with PCF improve the prediction performance by ∼20% at most compared with a single subgraph. In addition, various mediators enhance the richness of BPs information, which also makes different subgraphs may produce approximate prediction results. As shown in Table 6, for the subgraph DDLP and DDMLP, the result curves were similar due to the similarity of mediators.

The impact of direct prior information on Bridging-BPs

Bridging-BPs was designed to predict potential DTIs using indirect prior information, indicating that no known DTIs were involved in obtaining node structural features. We argued that Bridging-BPs that used only indirect priori information, might provide the lowest threshold for prediction indicators of Bridging-BPs with partial direct priori information. To illustrate this, we designed a comparative experiment that introduced direct prior

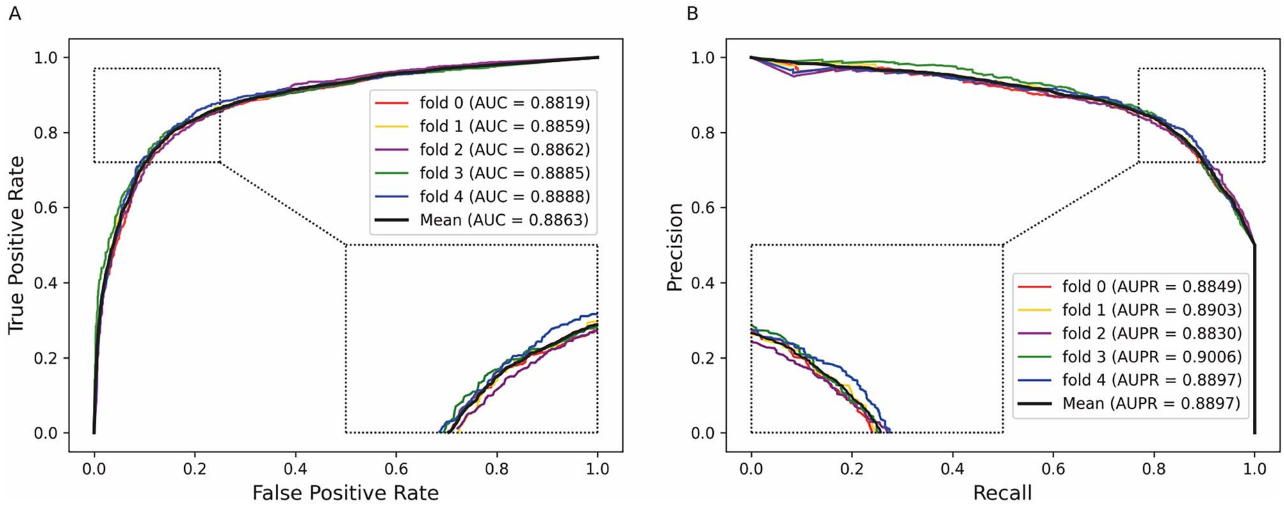

Figure 4. The ROC curves (A) and PR curves (B) of Bridging-BPs.

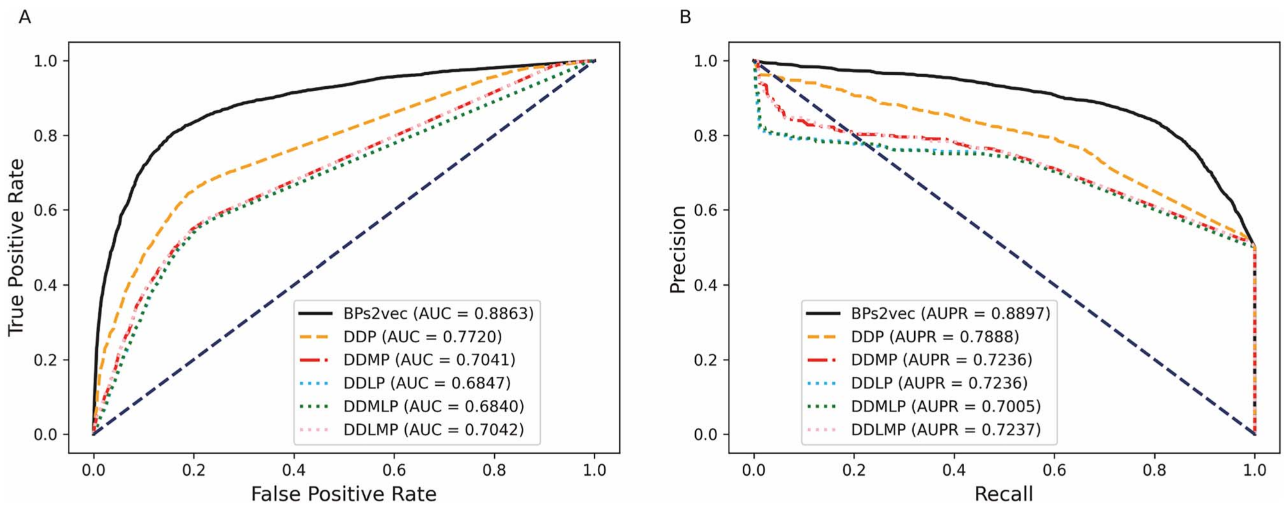

Figure 5. The ROC curves (A) and PR curves (B) of Bridging-BPs and subgraphs.

Table 6. The comparison results of the graph and its subgraphs

Type	Acc. (%)	Sen. (%)	Spec. (%)	Prec. (%)	MCC (%)	AUC (%)
DDP	72.69±0.83	65.3±0.98	80.09±1.23	76.64±1.16	45.9±1.70	77.20±0.98
DDMP	67.28±0.79	52.71±1.11	81.85±1.34	74.4±1.40	36.13±1.71	70.41±0.54
DDLP	66.62±0.89	52.25±1.05	80.99±1.27	73.33±1.43	34.71±1.91	68.47±0.87
DDMLP	66.63±0.89	52.27±0.98	80.99±1.35	73.34±1.48	34.72±1.92	68.40±0.74
DDLMP	67.30±0.85	52.67±1.22	81.92±1.35	74.46±1.44	36.18±1.81	70.42±0.60
Graph	81.98 ±0.45	83.16 ±0.63	80.8 ±0.88	81.25 ±0.66	63.98 ±0.88	88.63 ±0.27

information. Specifically, ensuring same parameters as the main experiment, we added positive samples in different proportions to a constructed bridging heterogeneous graph. It was worth noting that we added up to 80% positive samples into the graph. Experimental results are showed in Table 7 and Figure 6. The results indicate that, with adding 80% of positive data, Bridging-BPs can achieve up to AUC of 96.73% and AUPR of 96.80%. When direct prior information is lacking, Bridging-BPs can provide reasonable node structure information for DTs. Meanwhile, adding some direct prior information could effectively enrich the structure information of a graph and improve the prediction effect of Bridging-BPs.

Comparison of different NRL methods

One of the salient characteristics of BPs2vec was that BPs2vec take into account the requirements of downstream tasks. When sampling, node2vec only considers the depth-first and breadth-first search. This means that node2vec focuses on the randomness of walking, while ignoring those sequences that might need more attention in downstream prediction tasks. We compared

Table 7. The comparison of prediction results under different proportions of positive samples

Add	Acc. (%)	Sen. (%)	Spec. (%)	Prec. (%)	MCC (%)	AUC (%)
0.0	81.98±0.45	83.16±0.63	80.80±0.88	81.25±0.66	63.98±0.88	88.63±0.27
0.2	86.97±0.67	85.51±1.29	88.43±1.28	88.10±1.10	73.98±1.34	93.22±0.78
0.4	89.74±0.45	88.38±1.04	91.10±1.51	90.88±1.35	79.53±0.94	95.60±0.54
0.6	90.46±0.31	89.07±1.52	91.86±1.15	91.64±0.99	80.98±0.59	96.26±0.50
0.8	90.69±0.32	89.05±1.00	92.33±0.88	92.08±0.78	81.44±0.63	96.73±0.35

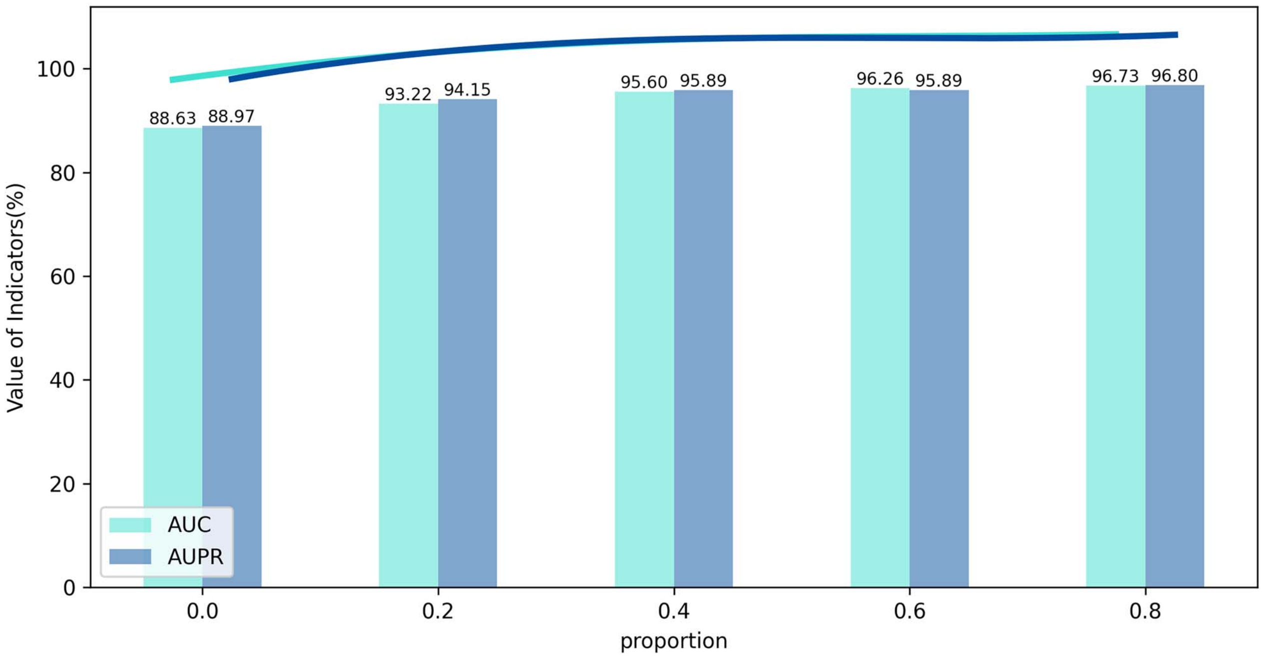

Figure 6. The changing trend of AUC and AUPR values under different introduction proportions.

prediction effects of BPs2vec and node2vec for DTIs with the same data set. In this comparison experiment, two hyper-parameter p and q in node2vec were set to 0.25 respectively. In addition, node2vec and BPs2vec were both set the default step size of 80. The results show that the prediction performance of BPs2vec was improved by ∼2%, which demonstrate that our improvement strategy is reasonable and effective.

Moreover, we compared BPs2vec with the end-to-end embedding method, Graph Convolutional Network (GCN) [34]. The results show that GCN has poor prediction indicators without direct prior information. The prediction effect of GCN is related to nodes own attributes. When little attribute information existing, GCN might perform poorly. Since no direct prior information involves in our model, the attribute features of DTs were not considered in the training process of GCN. The comparison results are shown in Table 8 and Figure 7.

Case study

Drug–targets screening

Here, we tested the generalization of Bridging-BPs in predicting novel DTIs. In addition to 5464 pairs of known DTIs and 5464 pairs of DTIs selected as negative samples, we applied post-training Bridging-BPs to predict other unknown DTIs and check the results through DrugBank and SuperTarget [35]. Specifically, we first utilized a random sequence to get as random DTs set as possible. Then, we randomly generated 100 pairs of new DTIs that are neither in the positive sample nor the negative sample, which are fed into the trained Bridging-BPs. After that, we rank the predicted results to obtain the top 20 DTIs predicted to be associated. The verification results are shown in Table 9. Verification results show that 16 of the 20 pairs of DTIs had been confirmed by DrugBank and SuperTarget.

Further exploring the prediction results, we find that the 20 pairs of unknown DTIs contain various types of target proteins, such as receptors, enzymes, transporters and carrier proteins. Among them, 16 pairs of DTs are directly identified by DrugBank as known DTIs, while proteins are identified as transporter, enzyme and carrier respectively in remaining four pairs. For example, p-glycoprotein 1, as a transporter of paclitaxel, becomes its specific target [36]. Due to their high specificity and high catalytic activity, UDP-glucuronosyltransferase 1A1 [37] and Cytochrome P450 1B1 can be used as targets

Table 8. The comparison results among BPs2vec, node2vec and GCN

Fold	Acc. (%)	Sen. (%)	Spec. (%)	Prec. (%)	MCC (%)	AUC (%)
BPs2vec	81.98±0.45	83.16±0.63	80.8±0.88	81.25±0.66	63.98±0.88	88.63±0.27
Node2vec	80.75±0.12	81.92±0.46	79.57±0.49	80.04±0.31	61.51±0.25	86.68±0.03
GCN	75.76±1.08	75.97±1.32	75.55±2.29	75.68±1.61	51.54±2.14	82.27±0.91

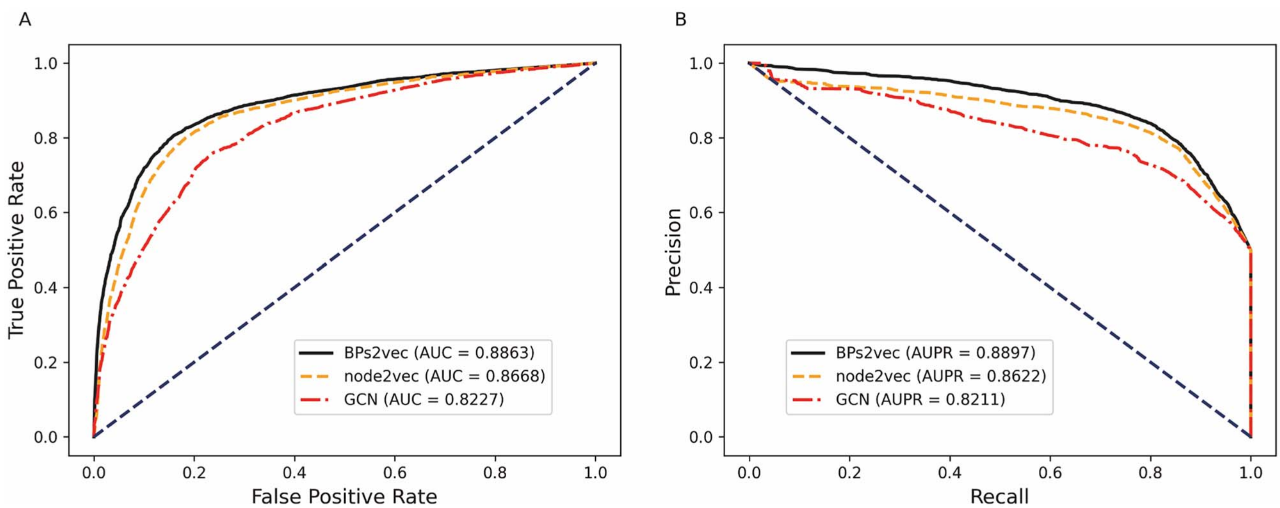

Figure 7. The ROC curves (A) and PR curves (B) of different NRL methods.

Table 9. The condition of candidate DTIs predicted by Bridging-BPs

Drug	Protein (STRING)	Type	Evidence
Verapamil	9606.ensp00000360423	Target	SuperTarget
Phenacetin	9606.ensp00000354612	Target	SuperTarget
Fluorouracil	9606.ensp00000315644	Target	SuperTarget
Methadone	9606.ensp00000394624	Target	SuperTarget
Meprobamate	9606.ensp00000274545	Target	SuperTarget
Prazosin	9606.ensp00000368766	Target	SuperTarget
Phenoxybenzamine	9606.ensp00000368766	Target	DrugBank
Paclitaxel	9606.ensp00000478255	Transporter	DrugBank
Naloxone	9606.ensp00000295897	Carrier	DrugBank
Spironolactone	9606.ensp00000274192	Target	SuperTarget
Furosemide	9606.ensp00000304845	Enzyme	DrugBank
Docetaxel	9606.ensp00000478561	Enzyme	DrugBank
Nitrazepam	9606.ensp00000469332	Target	SuperTarget
Ketoconazole	9606.ensp00000353820	Target	SuperTarget
Ibuprofen	9606.ensp00000385523	Target	SuperTarget
Estradiol	9606.ensp00000359285	Target	SuperTarget
Imipramine	9606.ensp00000219833	Target	SuperTarget
Rosiglitazone	9606.ensp00000310928	Target	SuperTarget
Quercetin	9606.ensp00000336528	Target	SuperTarget
Spironolactone	9606.ensp00000266376	Target	SuperTarget

of furosemide and docetaxel, respectively. Similarly, Albumin can serve as carriers for macromolecules of naloxone drugs.

Drug discovery of COVID-19

Coronavirus disease 2019 (COVID-19), caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), is an urgent global threat, with its rapid spread and high-mortality rate causing severe disruption [38]. The main clinical symptom of COVID-19 is pneumonia. But as the scale of infection increases, more and more clinical manifestations are being identified, including asymptomatic, acute respiratory distress syndrome, respiratory failure or multiple organ failure [39]. However, there are no specific antiviral drugs currently approved for the treatment of COVID-19. Due to the poly-pharmacological properties of drugs, an effective strategy is identifying therapeutic intervention with alternative medicines until a COVID-19 treatment is approved for production [40]. In this case study, we identified the primary targets of

Table 10. COVID-19 related targets of chloroquine, dexamethasone and ibuprofen

Drug	Target	Associations	probability
Chloroquine	Glutathione S-transferase A2	Inhibiter	0.80
Dexamethasone	Glucocorticoid receptor	Agonist	0.86
	Nuclear receptor subfamily 1 group I member 2	Agonist	0.87
Ibuprofen	Prostaglandin G/H synthase 2	Modulator	0.98
	Prostaglandin G/H synthase 1	Inhibitor	0.92
	Apoptosis regulator Bcl-2	Modulator	0.86
	Peroxisome proliferator-activated receptor gamma	Activator	0.84
	Protein S100-A7	Inducer	0.77
	Cystic fibrosis transmembrane conductance regulator	Inhibitor	0.80

three drugs (chloroquine, dexamethasone and ibuprofen) using Bridging-BPs, and examined their association with COVID-19 using DrugBank and related literatures. The prediction results are shown in Table 10.

Chloroquine is an antimalarial drug and it is also used for second line treatment for rheumatoid arthritis [41]. The potential drug–target relationship between chloroquine and Glutathione S-transferase A2 was early identified by investigators [42]. Moreover, the SARS coronavirus was found to be inhibited in cells culture by both chloroquine and hydroxychloroquine [43]. Indeed, molecular dynamics studies have been conducted to investigate the binding stability of chloroquine and Glutathione S-transferase A2 [44], which provides important insights into drug design for COVID-19 treatment. Although there are still some questions about the effectiveness of chloroquine treatment, clinical trials are ongoing.

In identifying drug targets for Dexamethasone, we validated two potential targets for COVID-19 at DrugBank: glucocorticoid receptor and nuclear receptor subfamily 1, group I member 2. Dexamethasone is an available glucocorticoid used to treat various inflammatory diseases, including bronchial asthma, as well as endocrine and rheumatic diseases [45]. In clinical experiments with patients with COVID-19, low doses of Dexamethasone have been shown to be effective in reducing mortality in severe cases, but not in mild cases [46].

Among the high-score targets associated with ibuprofen, six novel Coronavirus targets were found. These targets have been well documented in related studies [47–52]. In fact, ibuprofen has been thought to increase mortality, but studies have shown no significant difference in patients' health outcomes between the ibuprofen and non-ibuprofen groups. More research confirms this [53, 54]. The WHO has also relaxed restrictions on ibuprofen use. In addition, two types of clinical trials have been conducted.

Discussion and conclusion

In this paper, we first reviewed some classical methods of DTIs prediction and put forward our own views. Then, through analyzing defects of MAN, we explored a new way of constructing graph, and proposed an improved BPs-biased embedding algorithm BPs2vec. Compared with traditional embedding algorithms, BPs2vec developed more purposeful sampling strategies and considered contribution differences of subgraphs. We compared predictive effects of BPs2vec with other NRL methods (i.e. node2vec and GCN), and showed that BPs2vec had better prediction metrics. Besides, comprehensive experiments demonstrated the accuracy and robustness of our model without direct prior information. Case studies also showed that Bridging-BPs can predict novel DTIs more reliable and stable.

Moreover, we discussed the following issues. First, in Bridging-BPs, known DTIs were not involved in DTs feature acquisition, which has been experimentally proved to be effective. Second, in the selection of mediators, nodes that could link source nodes and destination nodes as much as possible should be selected in principle. Third, we chose up to three mediators to construct BPs. According to the weight calculation formula (7), when too many mediators were selected, the weight of the walking sequences of the corresponding subgraph will become lower. To some extent, we argued that such walking sequences had a lower predictive contribution to DTIs prediction.

Currently, Bridge-BPs has only been applied to DTIs prediction, whereas other associations, such as lncRNA-disease associations and miRNA-disease associations, have not been discussed. This will be a focus of our future work. Besides, since the current implementation of BPs2vec handles only heterogeneous graph data with the aim of obtaining structural features, in future, we plan to expand the functionality of Bridging-BPs to capture node features more precisely combined with their own properties.

Key Points

- Inspired by MAN, we innovatively build a bridging heterogeneous graph and use indirect prior information to speculate the potential DTIs in a non-end-to-end processing mode.
- A computable factor PCF is defined to measure the expected contribution value of each subgraph.
- An improved embedding algorithm BPs2vec based on Node2vec is proposed to learn the embedded features of nodes more tendentiously.

- Bridging-BPs is a complement to DTIs prediction and indirect prior information utilization.

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